

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	3
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to SQL Server and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in SQL Server to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

Results		Messages
	gender	revenue
1	Male	157890
2	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

Results		Messages
	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

Results Messages		
	item_purchased	Average Product Rating
1	Gloves	3.860000
2	Sandals	3.840000
3	Boots	3.820000
4	Hat	3.800000
5	Skirt	3.780000

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping

Results Messages		
	shipping_type	(No column name)
1	Express	60
2	Standard	58

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

Results Messages				
	subscription_status	total_customers	avg_spend	total_revenue
1	No	2847	59	170436
2	Yes	1053	59	62645

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

Results		Messages
	item_purchased	discount_rate
1	Hat	50.00000000000000
2	Sneakers	49.66000000000000
3	Coat	49.07000000000000
4	Sweater	48.17000000000000
5	Pants	47.37000000000000

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

Results		Messages
	customer_segment	Number of Customers
1	New	83
2	Returning	701
3	Loyal	3116

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

Results		Messages		
	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

Results		Messages	
	subscription_status	repeat_buyers	
1	Yes	958	
2	No	2518	

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

Results		Messages
	age_group	total_revenue
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

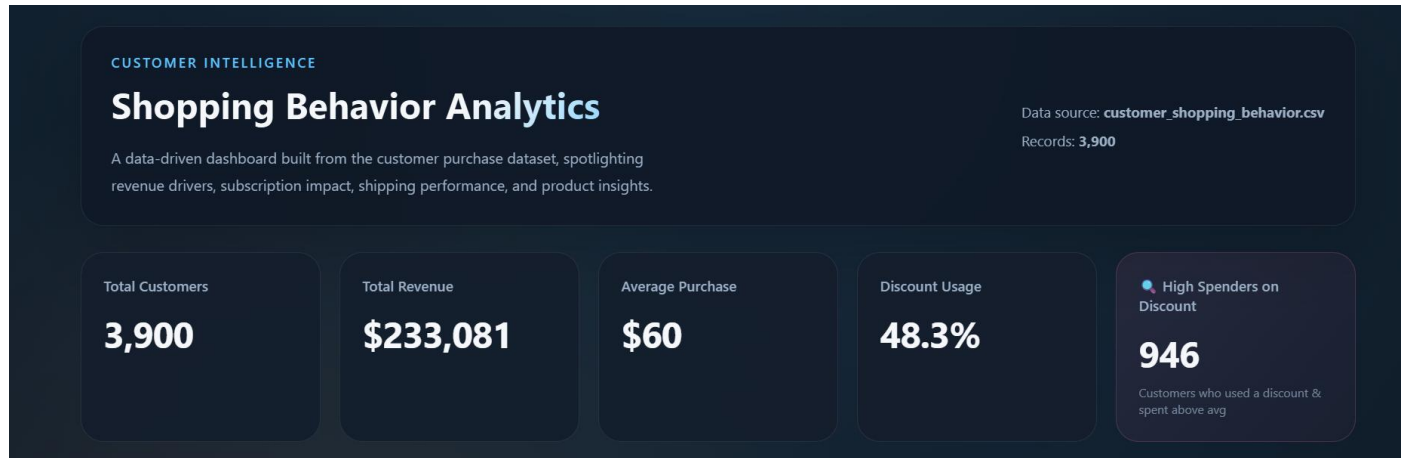
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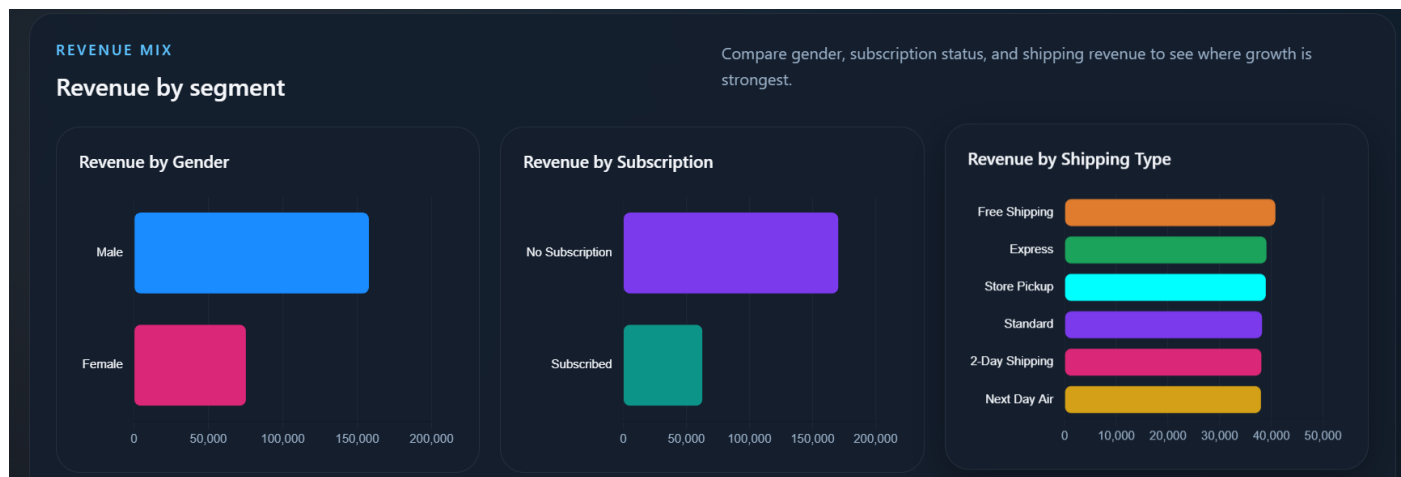
6.1: Home Page and KPI Overview of the Shopping Behavior Analytics Dashboard.



6.2: Revenue Analysis by Gender, Subscription Status, and Shipping Type.

This section analyzes revenue from three different perspectives:

- Revenue generated by gender
- Revenue generated by subscription status
- Revenue generated by different shipping methods



6.3: Revenue Distribution Across Different Age Groups.

The demographic dashboard analyzes customer spending across different age categories using both a bar chart and a donut chart:

- Customers aged **25–34** generated the highest revenue.
- Customers aged **35–44** represented the second-largest revenue segment.
- Revenue gradually declined for customers above 45 years.
- Customers aged **65+** contributed the least revenue.



6.4: Top Three Purchased Products Across Product Categories.

This visualization helps identify the best-performing products within each category, enabling businesses to optimize inventory planning, prioritize high-demand products, and design targeted promotional campaigns. The results indicate that Clothing remains the strongest product category, while Accessories, Footwear, and Outerwear also demonstrate consistent customer demand through their top-selling products.



6.5: Purchase Frequency, Payment Preferences, and Customer Segmentation.

This dashboard focuses on customer purchasing behavior by analyzing purchase frequency, preferred payment methods, and customer segmentation.

- Most customers purchase products every three months.
- PayPal and Credit Card are the most preferred payment methods.
- Loyal customers represent the largest customer segment.
- Returning customers contribute a significant portion of repeat purchases.

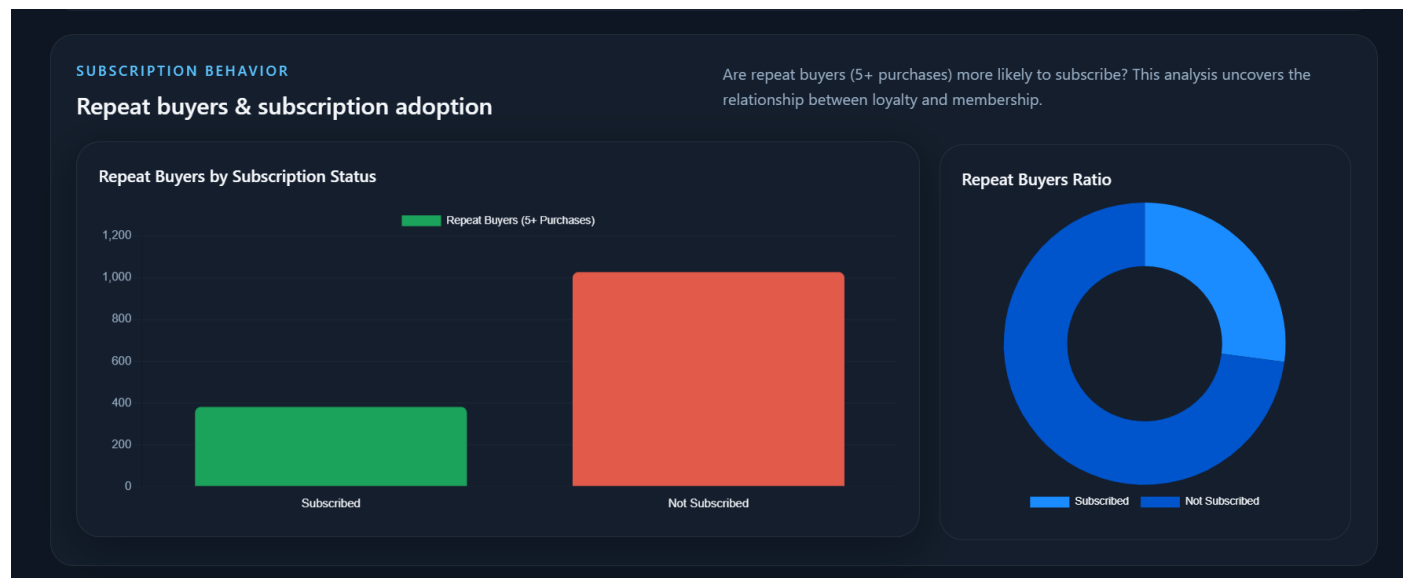


6.6: Repeat Buyer Analysis Based on Subscription Status.

This section investigates whether customers with subscriptions demonstrate stronger purchasing loyalty compared to non-subscribers.

The dashboard compares repeat buyers across subscription groups and visualizes their overall distribution.

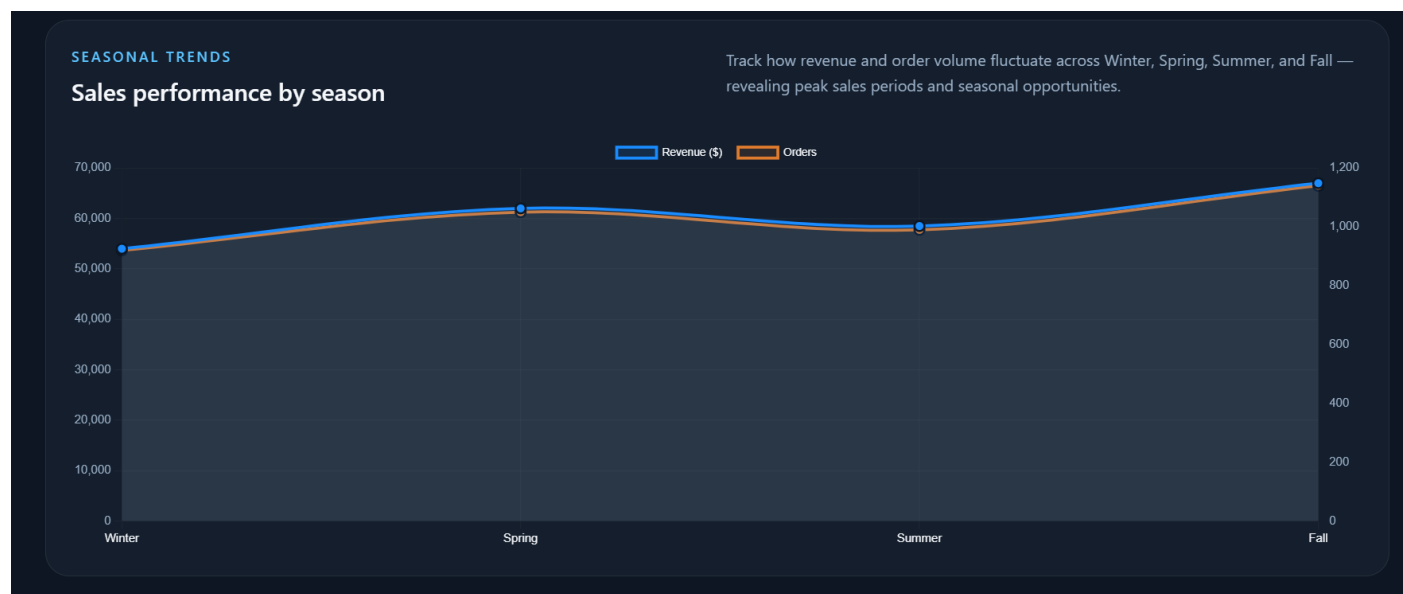
- Non-subscribed customers account for the majority of repeat purchases.
- Subscription adoption remains relatively low.
- A significant opportunity exists to increase subscription membership among repeat buyers.



6.7: Revenue and Order Trends Across Seasons.

Seasonal analysis evaluates changes in both revenue and order volume throughout Winter, Spring, Summer, and Fall.

- Fall generated the highest revenue and order volume.
- Spring also performed strongly.
- Summer showed a slight decline compared to Spring.
- Winter recorded the lowest overall sales.



7. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users